

Real-World AI-IoT Adversarial Problem Statements

Autonomous Vehicles (Transportation) – Traffic Sign and Sensor Spoofing



In real-world trials, inexpensive physical alterations (e.g. stickers or patches) on road signs can trick autonomous driving vision systems into “seeing” nonexistent commands or ignoring real ones ¹ ². Researchers showed that placing multicolored sticker patterns on Stop/Speed Limit signs caused top-brand self-driving cars to misinterpret them – leading to ignored stops, phantom braking, or speeding ¹ ². Once fooled, the car’s AI can perform unsafe maneuvers (e.g. accelerating through a stop sign), posing life-threatening risks. This problem is aggravated by the fact that modern AV safety features were not designed to handle adversarial sensor inputs ¹ ².

- **Industry:** Autonomous vehicles / smart transportation.
- **Why existing solutions are insufficient:** Current AV perception systems assume benign inputs and lack robust defenses against physical adversarial patterns. Commercial self-driving stacks (Tesla Autopilot, Waymo, etc.) have not been validated for such spoofing attacks ¹ ². Experts warn these new attacks expose “broken assumptions” in real-world AV vision systems and must be studied before deployment to ensure on-road safety ³ ⁴.

Healthcare Devices – AI Diagnostics and Monitoring Errors

Medical IoT devices (e.g. AI-driven diagnostics or patient monitors) can produce dangerous misreadings if their sensor inputs are manipulated. For example, adversarial attacks on patient data can cause AI algorithms to misdiagnose conditions or alter life-critical readings ⁵ ⁶. Published research

demonstrated a practical attack on an AI-controlled glucose-management system: by spoofing the Bluetooth link, attackers forced the system to report falsely high glucose levels, which could cause a diabetic patient to receive a fatal insulin overdose ⁶ . Similarly, small perturbations to ECG or imaging data could silently trick ML-based diagnostic tools. These failures directly threaten patient safety (delayed/missed alerts, wrong treatment decisions) and undermine trust in AI medical devices.

- **Industry:** Healthcare and medical devices.
- **Why existing solutions are insufficient:** Traditional device-security standards (like HIPAA compliance) do not address AI-specific threats ⁷ . Most FDA-approved AI/ML medical devices have not been evaluated against adversarial inputs ⁶ ⁸ . Conventional cybersecurity (firewalls, antivirus) cannot detect manipulated inputs that look normal to humans ⁹ . As one expert notes, “Traditional cybersecurity measures... are insufficient to counter adversarial AI attacks” ⁹ , so healthcare AI remains vulnerable to data spoofing with potentially fatal outcomes.

Industrial Automation – Robotics and Sensor Data Tampering

AI-enabled industrial robots and control systems can catastrophically fail if their sensors or training data are poisoned. For example, attackers could inject bad samples into an AI vision model’s training set, causing an assembly-line robot to misalign parts or ignore defects ¹⁰ . In lab tests, security teams showed that tiny, invisible patterns on objects caused robotic arms to misclassify items and drop them from height – directly risking worker injury ¹¹ ¹² . Poisoned data might also make robots “see” phantom parts or skip safety checks, resulting in faulty welding or collisions. These AI-specific attacks (model poisoning, adversarial vision inputs) combine cyber and physical hazards: a compromised robot arm could damage expensive equipment or hurt people.

- **Industry:** Manufacturing and industrial IoT.
- **Why existing solutions are insufficient:** Factories often deploy AI robots with little change management or ongoing security evaluation. Standard OT security focuses on network and patching, not on maliciously altered AI inputs ¹⁰ ¹¹ . Many industrial control systems still use legacy firmware with default credentials (see Amberwolf incidents ¹³). No off-the-shelf solution yet effectively defends against ML poisoning or crafted visual attacks in real time ¹⁰ ¹² , leaving a gap that could “disrupt production, compromise worker safety, [and] cause economic losses” if exploited.

Smart Grid and Infrastructure – Sensor Data Manipulation

Modern utilities rely on vast IoT sensor networks and AI analytics. However, attackers who spoof or corrupt sensor readings can make the grid “hallucinate” false conditions. A senior Analog Devices engineer warns that “**manipulated sensor data can lead to incorrect operational decisions, outages, or equipment damage**” in decentralized energy systems ¹⁴ . For instance, false readings from power meters or control substations could trick demand-forecasting AI into overloading lines or rerouting power unnecessarily. In turn, this might trip breakers or destabilize frequency control, risking blackouts. The human impact is severe: a cascading outage could cut power to hospitals and transportation (often noted in ICS incidents), endangering lives.

- **Industry:** Utilities and critical infrastructure (smart grids, energy).
- **Why existing solutions are insufficient:** The energy sector’s current security playbooks are primarily IT-centric (patching, firewalls) and do not cover AI data integrity ¹⁴ . Utility systems often

lack built-in safeguards for sensor data validation; data lakes and AI models assume trustworthy inputs. As experts note, securing these systems “requires hardware-based trust and secure-by-design architectures” beyond software patches ¹⁴, a shift only beginning to appear. Without AI-specific defenses (e.g. anomaly detectors tuned for adversarial tampering), smart grids remain exposed to these new risks.

Home/Facility Surveillance – Camera and Sensor Blinding

AI-driven security cameras and doorbell systems can be easily disabled by crafted inputs. In one case study, researchers used a common laser pointer to “blind” AI video doorbells: the bright coherent light reflecting inside the lens created patterns that confused the AI, making it report “*no motion*” even during a burglary ¹⁵. In effect, thieves could walk in view, and the system would neither record nor alert. This optical attack exploits a design flaw in the AI’s perception pipeline, not a software bug. Similar vulnerabilities exist: adversarial accessories (patches, clothing patterns) can fool face-recognition or motion-detection algorithms in surveillance cameras. The result is loss of situational awareness – homeowners or security staff get no alerts to wrongdoing, eroding trust in “smart” security systems.

- **Industry:** Consumer/home security and surveillance.
- **Why existing solutions are insufficient:** Commercial smart cameras were not designed to resist adversarial optical interference ¹⁵. Traditional countermeasures (better encryption, secure firmware) do not stop an attacker from physically shining a light into the lens. The Auburn study notes that fixing this requires “changes to algorithms and workflows” and even hardware redesign ¹⁶. In practice, no vendor offers built-in anti-laser or adversarial-attack protection today, so these devices remain vulnerable to simple physical spoofing.

Autonomous Drones (UAVs) – GPS and Sensor Spoofing

Unmanned aerial systems typically use AI for navigation and sensing; falsified inputs can cause crashes or hijacking. Drones rely on GPS for position data, and researchers warn that “**victim drones using civil GPS can be misdirected or even completely hijacked for malicious intents**” ¹⁷. In experiments, low-cost spoofing hardware has successfully sent fake GPS signals to UAVs, diverting them from their course or tricking them into landing. A hijacked survey drone might crash into people or critical assets, while military UAVs could be captured. Beyond GPS, adversaries can also jam or spoof onboard sensors (e.g. inertial or visual) to confuse AI autopilots. Such attacks can rapidly turn automation into a safety hazard, undermining confidence in drone operations.

- **Industry:** Aerospace, logistics, defense (UAVs).
- **Why existing solutions are insufficient:** Most commercial drones (and even some defense systems) use unencrypted GPS without spoof-detection. The cited review notes that GPS spoofing attacks “have serious implications” and are demonstrated with off-the-shelf gear ¹⁷. Anti-spoofing measures (e.g. multi-band GNSS or sensor fusion checks) are only now emerging and are not universally deployed. Until robust spoof-resistant navigation is standard, AI navigation remains vulnerable – and existing flight-control systems offer no complete defense against these stealthy input manipulations ¹⁷ ¹⁸.

Sources: Each statement above is drawn from public reports and research (whitepapers, news releases, or expert blogs) on AI/IoT vulnerabilities ¹ ¹⁹ ⁶ ¹⁰ ¹⁴ ¹⁵ ¹⁷ , illustrating real operational risks and gaps in current defenses.

¹ ² ³ ⁴ UC Irvine study shines headlights on consumer driverless vehicle safety deficiencies – UC Irvine News

<https://news.uci.edu/2025/03/04/uc-irvine-study-shines-headlights-on-consumer-driverless-vehicle-safety-deficiencies/>

⁵ ⁶ ⁷ ⁸ ⁹ ¹⁹ How adversarial AI threatens healthcare security systems

<https://www.paubox.com/blog/how-adversarial-ai-threatens-healthcare-security-systems>

¹⁰ ¹¹ ¹² ¹³ Cybersecurity Risks in AI-Powered Industrial Robots: The New Frontier of Digital-Physical Threats

<https://www.linkedin.com/pulse/cybersecurity-risks-ai-powered-industrial-robots-new-frontier-verma-1cfpc>

¹⁴ Smart grids are trying to modernize and attackers are treating it like an invitation - Help Net Security

<https://www.helpnetsecurity.com/2025/12/04/sonia-kumar-analog-devices-securing-smart-grids/>

¹⁵ ¹⁶ Assistant professor in CSSE exposes vulnerability in AI-powered security cameras, offers new defenses in development

<https://eng.auburn.edu/news/2025/10/assistant-professor-in-csse-exposes-vulnerability-in-ai-powered-security-cameraselopment.html>

¹⁷ ¹⁸ On GPS spoofing of aerial platforms: a review of threats, challenges, methodologies, and future research directions - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8114815/>